



Continuous Cell Separation Using Microfluidic-Based Cell Retention Device with Alternative Boosted Flow

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Received: 5 December 2019 / Accepted: 13 February 2020 /

Published online: 21 February 2020

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Abstract

The development of a continuous process for cell separation is growing rapidly due to the current trend of cost-effective manufacturing in biological industries. The continuous cell separation process has a significant reduction in capital equipment costs and facility size compared to the conventional batch process. In the study, a multi-layered microfluidic-based device integrated with the porous membranes was fabricated for continuous size-based isolation of the cells based on the mechanism of restrictive cross-flow filtration, allowing the biological sample entered in a single inlet of the device and separated into two outlet streams. One stream which contained the cells returned back to the original sample fluid, while another stream with conditioned medium only was collected for later applications. The membrane fouling issue was overcome by introducing the alternative flow rate consisted of a set of higher and lower flows. The device integrated with the controllable flow restriction allows to increase the permeate flow rate, and alternative boosted flow demonstrates the high permeate flow rate (0.3 mL/min), high cell viability (> 98%), and increase of cell concentration (48%). As a result, we believe that the microfluidic-based continuous cell separation system is a promising tool for downstream bioprocess.

Keywords Continuous bioprocess · Cell separation · Microfluidics · Cross-flow filtration · Antifouling

Introduction

Continuous bioprocessing technology has been developed for several decades. Recently, it has received wide attention because of manufacturing cost reduction, productivity increase, and product quality uniformity compared with the conventional batch process [1–3]. This biomanufacturing

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process including upstream production and downstream purification has been used to continuously produce high-value biological products, including therapeutic monoclonal antibodies [4, 5], viral vaccines [6], and recombinant human enzymes [7]. Continuous cell culture through the perfusion process plays a critical role that ensures high cell viability and high productivity. This is because in the perfusion culture system, fresh medium is continuously pumped into a bioreactor which contains cells and medium to maintain the cell concentration. The conditioned medium-containing waste and products are collected at the permeate flow outlet, while the cells are recycled back to the bioreactor after passing the cell retention device during cell culture to reduce the capital costs [8].

The cell retention device is analogous to the separation device that separates the cells from the medium based on the mechanism of centrifugation, gravitational settling, membrane filtration, ultrasonic separation, dielectrophoretic separation, or inertial separation [9–11]. Among these separation techniques, membrane-based cross-flow (tangential flow) filtration is commonly used over other techniques due to its simplicity, easy adaption and operation, high filtration capacity, high purity, and low operating costs without additional development work [9]. However, the major drawback that limits the development of this technology is membrane fouling which deteriorates the membrane permeability, leading to decrease in the permeate flow rate and recovery efficiency. Most importantly, the stuck cells on the membrane may reduce the overall cell viability due to the high shear stress close to the porous side [12, 13]. Therefore, effective ways to remove membrane fouling are essential in order to maintain high product yield of the membranes and prolong membrane lifetime. The common methods for reducing membrane fouling are cross flashing, backwashing, vibration, and backpulsing [14, 15]. Nevertheless, additional process steps that increase the process complexity and equipment which increase extra costs in the whole process are still necessary to prevent the membrane fouling.

In recent years, microfluidic-based approaches for continuous cell separation were developed for a high flow rate scale (mL/min per single device) [16]. These novel microfluidic separators have reached significant achievements such as precise separation of particles from suspension, low manufacturing cost, small footprint, and enhancement of product quality [17–20]. However, the separation throughput is relatively low when a single device is integrated into the system. Although the throughput can be increased by upscaling through parallelization or stacking that eventually apply for industrial applications, this arrangement increases the system complexity that retains the development of this novel process [21].

In the study, we demonstrated a controllable microfluidic-based cell retention device based on restrictive cross-flow filtration and alternative boosted flow for continuous cell separation. The device integrated with porous membranes so that the cells can be selectively isolated from the mainstream with minimum membrane fouling. During the process of cell separation, two sets of flow rates (the higher and the lower) can be alternatively applied by a computer-controlled peristaltic pump. The sample flow with a higher flow rate is utilized to flush cell clogging on the membranes, while the lower flow rate focuses on the enhancement of cell separation efficiency. The capabilities of the cell separation system are characterized by orchestrating the flow resistance, flow rate, and flow modes. Improvements in the device performance in terms of permeate flow rate, cell viability, cell concentration, and membrane fouling are subsequently demonstrated.

Principle

The mechanism of this study to retaining the particles (or cells) in the mainstream is size-based cross-flow filtration (Fig. 1a). The sample flow is introduced into the microfluidic-based

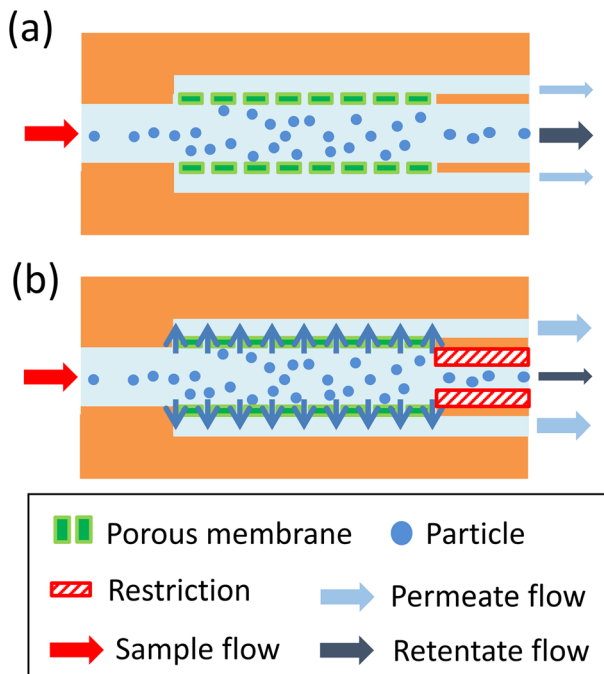


Fig. 1 **a** The schematic of continuous separation mechanism using cross-flow filtration to retain the particles in the retentate flow, while the filtrate is collected into the permeate flow. **b** The separation efficiency can be improved by increasing the resistance in the retentate flow

retention device, and the large size of the particles retains in the retentate flow, while the filtrate penetrates across the porous membrane into the permeate flow. According to Darcy's law, the separation efficiency (or particle concentration) can be enhanced by either increasing the retention time of the sample flow or the surface area of the porous membrane in the retention device. It is applicable if the size of the retention device is sufficiently large to encapsulate the larger porous membrane. However, if the overall size of the processing setup is critical, increasing the size of the porous membrane is not an optimized choice. Unlike the conventional cross-flow filtration, therefore, the separation efficiency can be controlled by simply adding a moderate resistance in the retentate flow, allowing to increase the internal pressure of the channel to drive more filtrate to penetrate across the membrane (Fig. 1b). Neutrally buoyant particles or cells suspended in a fluid can still flow through a straight microchannel without losing their moments for moving. In other words, the porous membrane integrated into the cell retention device is able to separate the cells from the medium and keeps them retained in the cell separation system, allowing to increase the efficiency of size-based separation.

Experimental Methods

Design and Fabrication of Cell Retention Device

The cell retention device was designed by using Solid Edge 2D software (ST9, Siemens PLM Software) and illustrated in Fig. 2 a. The device is composed of a total of 6 layers

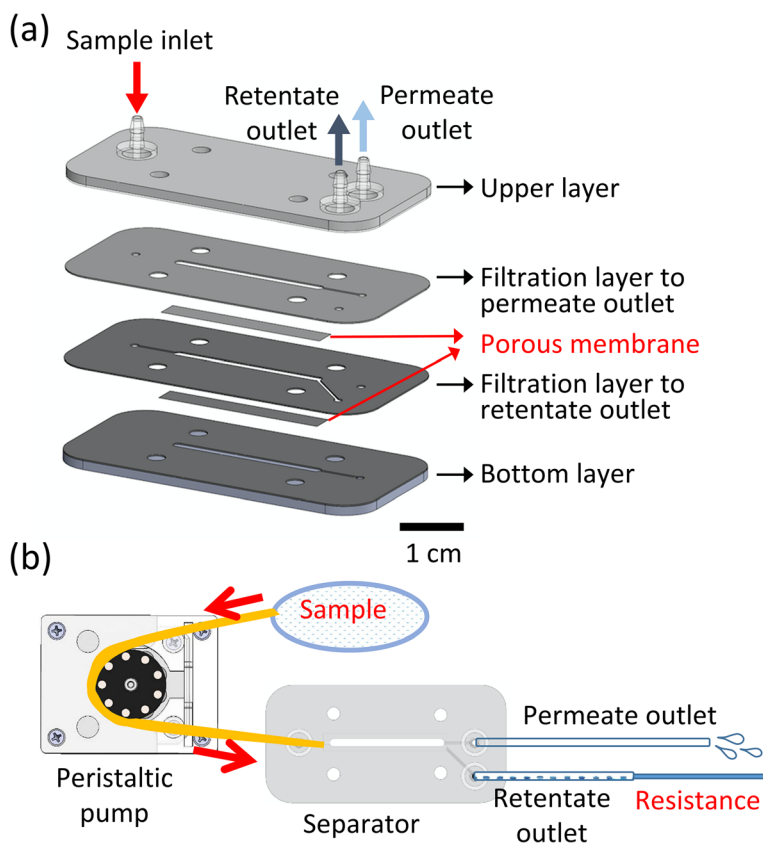


Fig. 2 **a** Schematic diagram of the cell retention device. Two layers of porous membranes are integrated into the device to increase separation efficiency. The sample mixture can be introduced into the device and separated into retentate and permeate streams through cross-flow filtration. **b** The schematic of system setup to enhance the separation efficiency by integrating the resistance in the outlet of the retentate

including 2 layers of the porous membrane. Sample flow can be introduced into the inlet of the device, directed into the channel layer (between two porous membranes) that retains the large size of particles, and collected toward the retentate outlet. The channel layer was designed with 0.25 mm in height and 1 mm in width. The penetrated filtrates that flow across the membrane are collected toward the permeate outlet. Each plastic layer was prelaminated with biocompatible adhesive tapes (Cat. No. 9122, 3 M Company) using a seam roller and cut by a laser system (PLS4.75, Universal Laser System) [22]. First, the 2D sketches were sent to the ULS Control Panel which was connected to the laser system, and the cutting alignment and cutting protocol (e.g., laser power, cutting speed) were set depending on the property and thickness of the materials. The porous membrane (5 μ m in pore size, polyethylene terephthalate, Pall) was cut in the size of 5 $\times$ 20 mm (width $\times$ length) using the laser cutter. Finally, each layer was aligned by alignment gigs and combined together to construct a 3D device. Polycarbonate tube adaptors (BDMR210–9, Nordson MEDICAL) were inserted in the inlet and outlet connections of the device and fixed with the epoxy glue.

System Setup

The flow of the separation system is driven by a peristaltic pump and showed in Fig. 2 b. The peristaltic pump used in this study was constructed by integrating a 4-roller pump head (Runze Fluid) on a stepper motor (DMX-K-DRV-17, Arcus Technology) and controlled by using a computer-based software (Arcus Technology). The system was set up by integrating the separator after the peristaltic pump to separate the mixture of the sample withdrawn from the sample reservoir. The silicone tubes (EW-95802-02, Cole-Parmer) with 1/16 in. inner diameter (ID) were connected to the tube adaptors on the device to direct the flow into permeate and retentate flows. Different length of the polyetheretherketone (PEEK) tube (IDEX Health & Science) was connected after the retentate outlet as the resistance of the flow. The flow rate of collected permeate and retentate flows was calculated by measuring the collected volume at a specific time. For measurement of separation performance, the neutral microparticles (polyester toner powder, Formujet) with the size ranged from 10 to 20 μm were added in the deionized (DI) water as the sample. For the experimental setup used in the cell separation, all of the devices, tubes, and connectors were sterilized using 5% of hydrogen peroxide (H_2O_2) in DI water, washed with DI water, and UV irradiated overnight in the cell culture hood to eliminate the remaining H_2O_2 in the system [23].

Image Acquisition and Analysis

The images of trapped particles on the membrane were captured under an inverted microscope (Eclipse TS100 with C-SHG light source, Nikon) connected to a digital camera system (LeadView 2800AM-FL, Leader Scientific). All images were analyzed using ImageJ software (NIH) to obtain information of particles trapped in the retention device at different time points within a specific region of interest (ROI).

Cell Culture

The epithelial lung carcinoma A549 cells (gift from Prof. Shang-Hsiu Hu, National Tsing Hua University) were maintained in 89% high-glucose Dulbecco's modified Eagle medium (DMEM, GE Healthcare) with sodium pyruvate (Hyclone, GE Healthcare), 10% cosmic calf serum (Hyclone, GE Healthcare), and 1% Penicillin/Streptomycin (P/S, GE Healthcare). The A549 cells were cultured in the humidified incubator at 37 °C with 5% CO_2 . High Five cells (gift from Prof. Yu-Chen Hu, National Tsing Hua University) were cultured in Sf-900 II serum-free medium (Thermo Fisher Scientific) containing 1% of P/S and culture in the humidified incubator at 27 °C.

Cell Viability and Number Analysis

Live and dead cell viability test is an indispensable tool to quantify the number of live and dead cells in cell biology research, providing a two-color fluorescence method to examine the cell viability. In the study, the live and dead cell viability assay (K502–100, BioVision) was applied to prove the feasibility of the continuous cell separation system. Each assay containing 2 μL of live cell staining dye (measured at Ex/Em = 485/530 nm) and 1 μL of dead cell staining dye (measured at Ex/Em = 495/635 nm) were added into 1 mL of sterilized phosphate-buffered saline (PBS, GE Healthcare) as a staining solution stock. Afterward, 100 μL of the sample

cells with the medium was drawn out from the bioreactor and placed in a 96-well plate. Added 100 μL of the prepared staining solution to the well and placed it into an incubator for 30 min at 37 $^{\circ}\text{C}$. After incubation, live and dead staining of the sample cells were examined under CELENA® S Digital Imaging System (Logos Biosystems), and viability and number were calculated by counting the number of live cells (green color) and dead cells (red color).

Setup of Continuous Cell Separation System

To ensure the mass balance during the continuous cell separation process, the flow rates of input and output were adjusted to be equally the same. The input was the fresh cell culture medium, while the output referred to the clarified culture medium. The initial cell stock was cultured with 20 mL of initial volume and 2×10^5 cells/mL of cell concentration in the centrifuge tube (Corning). The stir bar was used and driven by a stir plate at 60 rpm to maintain the suspension culture. The system was installed with two pumps: pump #1 (P1) was responsible for introducing fresh medium into the cell stock and pump #2 (P2) was used to withdraw the cell stock into the cell retention device for cell separation. The flow rate of P1 was initially set to 0.42 mL/min and adjusted with time to maintain the same volume of the cell stock. The flow rate of P2 was set at 0.42 and 0.9 mL/min periodically with frequency of 0.08 Hz. The cell viability was verified using the live and dead cell viability test at each time point during the continuous cell separation process.

Results and Discussion

Performance of the Separator by Adding Different Resistances

We first tested the performance of the cell retention device by adding different PEEK tube as resistances to enforce more fluid to penetrate through the porous membrane. Based on Poiseuille's law, the resistance (R) can be described as follows in Equation (1).

$$R = \frac{8\mu L}{\pi r^4} \quad (1)$$

where μ is the viscosity of the fluid, L is the length of the tube, and the r is the radius of the tube. The filtrate flux is proportional to the tube length and inversely proportional to the fourth power of the tube radius. Since the flow rate of the retentate flow is determined by the length and the radius of the inserted tube, the tube length from 4 to 10 cm was selected and the tube radius of 0.25 mm and 0.75 mm-inner diameter were chosen to verify the performance of the cell retention device. As shown in Fig. 3 a, the permeate flow rate linearly increases with the length of the tube inserted after the retentate flow for both two different tube radius. Compared with the results with different tube radius, the flow rate for 0.25 mm-ID setting is higher than the device integrated with 0.75 mm-ID tube. These experimental results are identically corresponding to the statement described previously, suggesting that the increase of resistance is able to enhance the separation efficiency. It should be noted that the flow rate of retentate flow remains the same for all resistance adding after the outlet of the retentate flow because the peristaltic pump is a positive displacement pump that can drive the sample flow into the separator without generating backflow. Nevertheless, the internal pressure in the channel dramatically increases when the tube radius becomes relatively small.

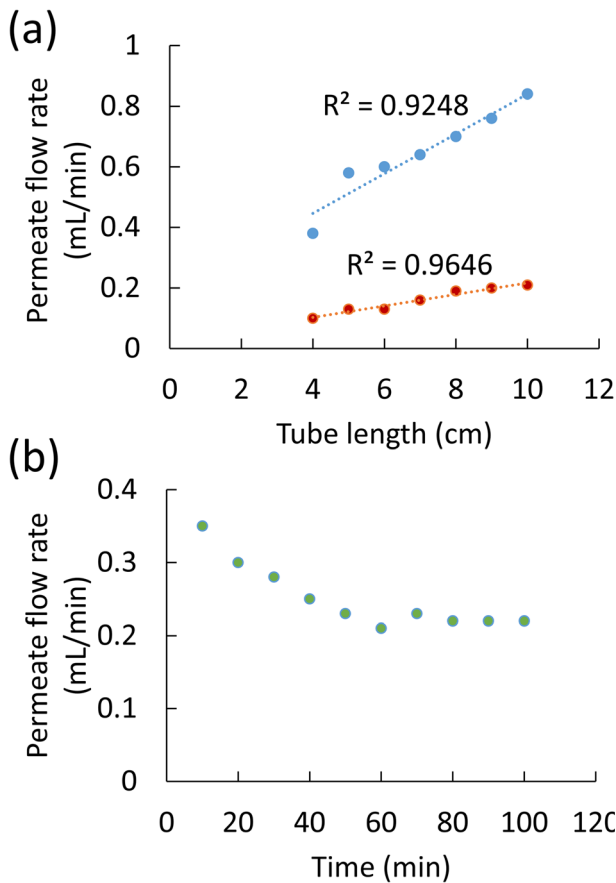


Fig. 3 **a** Comparison results of permeate flow rate using 0.75 mm-ID tube (blue dots) and 0.25 mm-ID tube (red dots) as the resistance with different tube lengths. **b** The permeate flow rate results when using 0.25 mm-ID tube as the resistance at different times. All samples were performed at the sample flow velocity of 244 mm/s

Membrane fouling is considered a major drawback for all types of membrane separation that could reduce the separation efficiency, increase the maintenance cost, and may stop the operation process [15]. To estimate the fouling time of the cell retention device, the resulting flow rate of the penetrate was obtained at different time points. As shown in Fig. 3 b, the permeate flow rate reaches to almost constant (0.22 mL/min) after 60 min of separation. The separation efficiency drops 37% compared with the initial separation at 10 min.

Enhancement of Separation Efficiency Using Alternative Boosted Flow

The increase of separation efficiency and reduction of maintenance costs (avoid to replace a new separator) are major concerns for the continuous cell separation process. To achieve these goals, we introduced a new strategy by alternatively tuning the high and low flow rate of sample flow to remove the membrane fouling. When the amount of collected filtrate reduces due to the accumulation of the particles on the membrane, the flow rate controlled by the computer starts to initiate the washing protocol to wash the particles from the membrane. The washing flow rate should be higher than the regular separation flow rate at a short period of

time to prevent the damage of the cells due to the high shear stress. Figure 4 a shows the changes in permeate flow rate when the alternative flow rate is applied. Initially, the flow rate can reach to the maximum (#2) due to the few particles accumulated on the membrane. The separation efficiency drops to the minimum (#3) after 10 min at the flow velocity of 244 mm/s. Next, the flow velocity is increased to 1910 mm/s to tangentially remove the particles from the membrane after 10 min (#1). Repeatedly, the separation efficiency is boosted every time when the high flow velocity is applied. The retention device, therefore, can be verified by observing

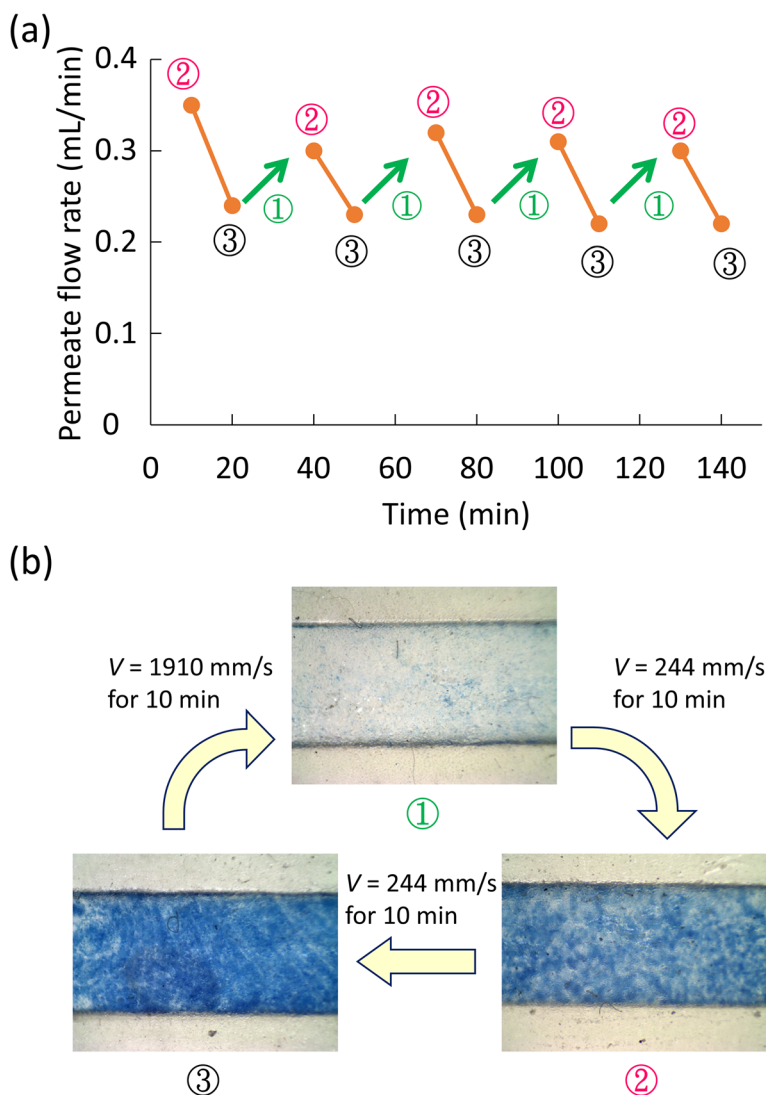


Fig. 4 **a** The flow rate changes of filtrate at different time points. The green arrow indicates the boosted flow rate process. #1 indicates the time point when the particles are removed. #2 indicates the time point when the filtration efficiency reaches the maximum. #3 indicates the time point when the filtration efficiency reaches to the minimum. **b** The images of the particles accumulated in the cell retention device at different time points. The normal separation flow velocity applied in this system is 244 mm/s, while the boosted velocity is 1910 mm/s

the accumulation of the particles on the membrane to ensure the consistent results at different separation time points. Figure 4 b demonstrates the increase of the flow velocity to 1910 mm/s for 10 min can dramatically remove most of the clogging on the membranes and maintain high filtration efficiency, suggesting an optimal approach to achieve continuous cell separation.

Evaluation of Cell Separation Using Cell Retention Device

The issue of cell clogging has become the principal impediment for the current membrane-based micro-separation technology which is widely used in the biomedical and pharmaceutical industry [24]. In order to solve the issue of cell clogging inside the flowing channel of the cell retention device, the alternative flow approach with two different flow rates is applied for continuous cell separation. The priority to determine optimal flow rate settings for continuous cell separation is to obtain high cell viability and reduce the cell clogging inside the flow channel of the retention device, leading to higher cell concentration at the retentate outlet. To further ensure the cell retention device can be used to perform the cell separation, the cell viability, the flow rate, separated cell concentration were evaluated. The sample flow consists of A549 cells with cell density of 2×10^5 cells/mL was introduced into the inlet of the cell retention device by using a peristaltic pump. Following the same cross-flow mechanism, the cell retention device allowed A549 cells separated from the medium, resulting in the outlet of permeate flow contains only conditioned medium and the outlet of retentate flow contains A549 cells were harvested. The total cell number in the suspension of the cell retention device was evaluated to investigate the number of cells clogged on the membrane under different input flow rates. Figure 5 a demonstrates that 2×10^5 cells clog on the membrane at the flow rate of 0.42 mL/min, suggesting that the low flow rate was not sufficient to carry the cells to the retentate flow and cause most of the cells to settle on the membrane. When the flow rate increased to 1.8 mL/min, the cell clogging reduces 74% to 5.3×10^4 cells due to less flow directed to the cross-flow direction. In order to reduce the cells clogging in the flow channel of the device and enhance medium collection efficiency, we applied two different flow modes at alternative flow rates by using a computer-controlled peristaltic pump. The separation mode (S-mode) indicates the cell separation under a lower flow rate, while the anti-clogging mode (A-mode) represents the fouling removal under a higher flow rate. When the S/A modes (flow rate = 0.42/0.9 mL/min, frequency = 0.08 Hz) were applied, the number of cells clogged on the membrane reduced to 2.7×10^4 cells equivalent to 87% of clogging compared to using only 0.42 mL/min of flow rate. The clogging can even drop to 97% with 6.7×10^3 cells trapped on the membrane when different S/A modes (flow rate = 0.9/1.2 mL/min, frequency = 0.08 Hz) were used.

Although the cell clogging issue can be mitigated by increasing the flow rate or applying the S/A modes, the cell viability is an important factor to determine which operation flow rate is suitable for cell retention. As shown in Fig. 5 b, the cell viability reached 99% at the flow rate of 0.42 mL/min. However, the low rate is not sufficient to maintain the high medium separation efficiency due to the clogging of the cells. When the input flow rate increased to 1.8 mL/min, the cell viability dropped to 81% as expected because of high shear stress. Interestingly, when using the S/A modes, the cells were able to maintain high cell viability up to 99% for S/A flow rate equal to 0.42/0.9 mL/min and 95% cell viability for S/A flow rate equal to 0.9/1.2 mL/min. This is a significant improvement for on-site cell separation, especially for the small scale microfluidic-based device. We further examined the concentration of purified cells in the retentate flow to identify the purification capability of the cell retention device. The results show that the cell concentration ($\sim 1.91 \times 10^5$ cells/mL) was lower

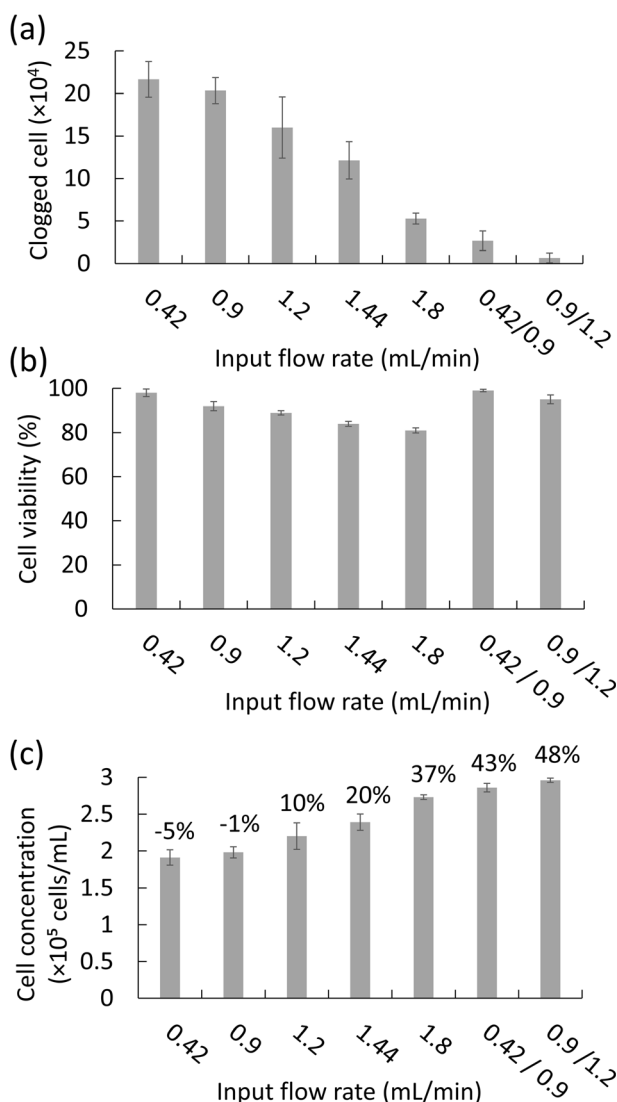


Fig. 5 The A549 cells with an initial cell concentration of 2×10^5 cells/mL were processed by the retention device at different flow rates. **a** The amount of the cells clogged on the membrane at different flow rates. **b** The cell viability when different flow rate was applied. **c** Cell concentration in the retentate flow at different flow rates. The increased cell concentration percentage is indicated on each data

than the initial concentration at the flow rate of 0.42 mL/min (Fig. 5c). This can be explained when some cells were trapped on the porous membrane, resulting in lower cell concentration in the retentate flow. The input flow rate was then subsequently increased, yielding 2.73×10^5 cells/mL of cell concentration at the flow rate equal to 1.8 mL/min. This value is 37% higher than the cell concentration in the original flow. We also observed the cell concentration increased by 43 and 48% when the S/A modes were applied at the flow rate of 0.42/0.9 and 0.9/1.2 mL/min, respectively. Based on these results, therefore, we chose the S/A modes with the flow rate of 0.42/0.9 mL/min for continuous cell separation.

Continuous Cell Separation

The conditioned medium obtained from the continuous cell separation process is assessed for biological products, such as monoclonal antibody [25]. Due to the permeation of conditioned medium from the separation system, therefore, the fresh medium is essential to be added back to the original cell stock to balance the total mass flow and prevent the accumulation of the waste in the original stock that leads to the low production rate [26]. To fulfill these requirements, a cell culture system connected to the cell retention device was established to continuously separate the cells from the conditioned medium. A fresh medium feed was pumped into the cell stock to keep the same mass flow and prevent the exhaustion of the medium (Fig. 6a). The High Five cells were chosen in this study due to their nature and high viability in suspension culture so that continuous cell separation can be achieved. The High Five cells with 2×10^5 cells/mL of cell density were continuously processed by adapting the S/A modes at the flow rates of 0.42/0.9 mL/min. As shown in Fig. 6b, the cell viability remains 98% after 10 h and 86% after 24 h of the continuous cell separation process and eventually dropped to 79% when cultured for 32 h. The total volume of permeate was collected to be about 600 mL, suggesting that this continuous separation system was successfully able to

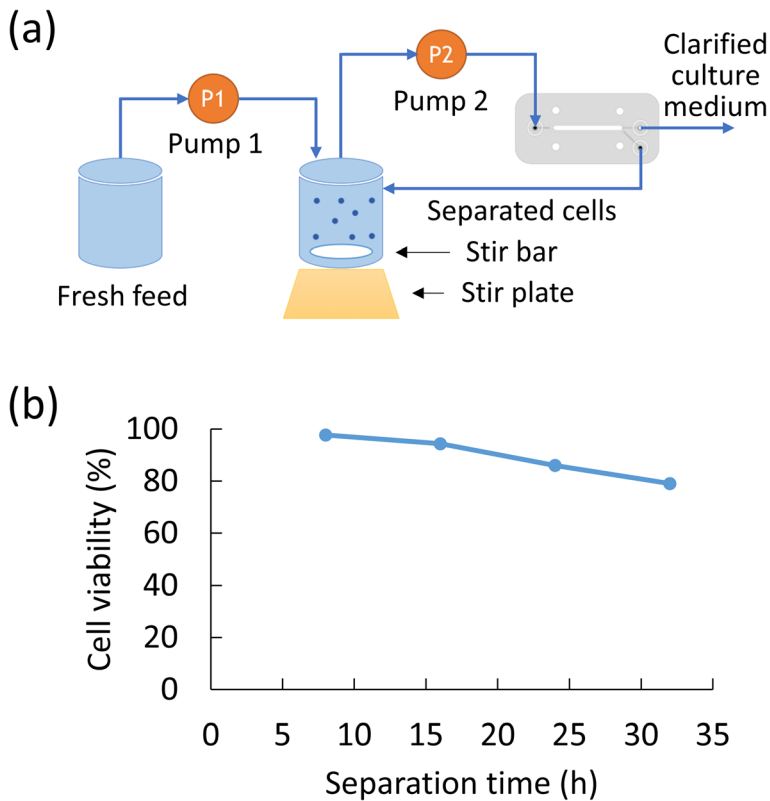


Fig. 6 **a** Schematic of continuous cell separation system with a combination of cell stock, cell retention device and fresh medium reservoir. Separated cells recycle back to the cell stock, while the clarified culture medium is collected for further use. The fresh medium is pumped into the cell stock to prevent the loss of the whole mass flow. **b** Cell viability measured at different time points when the continuous cell separation process is applied with the S/A modes at the flow rates of 0.42/0.9 mL/min ($N = 3$)

retain the cells and collect the conditioned medium on a large scale. Compared to the conventional membrane-based retention device, the major reasons of the retention device in this work are: (1) much smaller surface area of membrane in microfluidic-based device; (2) periodic S/A cell perfusion mode that maintains high cell separation efficiency; and (3) restrictive retentate flow which allows more permeate flow cross the membrane. Although the cell viability was examined lower after 24 h of the continuous separation process, the other factors to reduce the cell viability should be further verified, such as shear stress in the culture system.

Conclusion

In conclusion, we have developed a microfluidic-based cell retention device to continuously separate the cells and conditioned medium based on the restrictive cross-flow filtration combined with the alternative flow. This method enables the collection of conditioned medium for downstream recombinant protein purification, while simultaneously removing the membrane fouling to avoid cell clogging in the system. The resulting continuous separation process makes it possible to successfully maintain the cell viability for long-term cell culture – a critical bioprocess toward automation of protein purification or cell amplification.

Funding Information This work was supported by the Ministry of Science and Technology (MOST) in Taiwan under grant 107-2221-E-007-041-MY3.

Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

Informed Consent Informed consent was obtained from all participants included in the study.

References

1. Zydney, A. L. (2015). Perspectives on integrated continuous bioprocessing—Opportunities and challenges. *Current Opinion in Chemical Engineering*, 10, 8–13.
2. Zydney, A. L. (2016). Continuous downstream processing for high value biological products: A review. *Biotechnology and Bioengineering*, 113(3), 465–475.
3. Fan, J., Luo, J., & Wan, Y. (2017). Membrane chromatography for fast enzyme purification, immobilization and catalysis: A renewable biocatalytic membrane. *Journal of Membrane Science*, 538, 68–76.
4. Shukla, A. A., Wolfe, L. S., Mostafa, S. S., & Norman, C. (2017). Evolving trends in mAb production processes. *Bioengineering & Translational Medicine*, 2, 58–69.
5. Pollock, J., Coffman, J., Ho, S. V., & Farid, S. S. (2017). Integrated continuous bioprocessing: Economic, operational, and environmental feasibility for clinical and commercial antibody manufacture. *Biotechnology Progress*, 33, 854–866.
6. Gutiérrez-Granados, S., Gòdia, F., & Cervera, L. (2018). Continuous manufacturing of viral particles. *Current Opinion in Chemical Engineering*, 22, 107–114.
7. Warikoo, V., Godawat, R., Brower, K., Jain, S., Cummings, D., Simons, E., Johnson, T., Walther, J., Yu, M., Wright, B., McLarty, J., Karey, K. P., Hwang, C., Zhou, W., Riske, F., & Konstantinov, K. (2012). Integrated continuous production of recombinant therapeutic proteins. *Biotechnology and Bioengineering*, 109, 3018–3029.
8. Fisher, A. C., Kamga, M. H., Agarabi, C., Brorson, K., Lee, S. L., & Yoon, S. (2019). The current scientific and regulatory landscape in advancing integrated continuous biopharmaceutical manufacturing. *Trends in Biotechnology*, 37, 253–267.

9. Castilho, L. R., & Medronho, R. A. (2002). *Tools and Applications of Biochemical Engineering Science* (pp. 129–169). Berlin, Heidelberg: Springer Berlin Heidelberg.
10. Kwon, T., Prentice, H., Oliveira, J. D., Madziva, N., Warkiani, M. E., Hamel, J.-F. P., & Han, J. (2017). Microfluidic cell retention device for perfusion of mammalian suspension culture. *Scientific Reports*, 7, 6703.
11. Lee, D., Lee, Y., Choi, S. S., Lee, S. H., Kim, K. W., & Lee, Y. (2019). Effect of membrane property and feed water organic matter quality on long-term performance of the gravity-driven membrane filtration process. *Environmental Science and Pollution Research*, 26, 1152–1162.
12. van Reis, R., & Zydney, A. (2001). Membrane separations in biotechnology. *Current Opinion in Biotechnology*, 12, 208–211.
13. Radoniqi, F., Zhang, H., Bardliving, C. L., Shamlou, P., & Coffman, J. (2018). Computational fluid dynamic modeling of alternating tangential flow filtration for perfusion cell culture. *Biotechnology and Bioengineering*, 115, 2751–2759.
14. Karst, D. J., Serra, E., Villiger, T. K., Soos, M., & Morbidelli, M. (2016). Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes. *Biochemical Engineering Journal*, 110, 17–26.
15. Gao, Y., Qin, J., Wang, Z., & Østerhus, S. W. (2019). Backpulsing technology applied in MF and UF processes for membrane fouling mitigation: A review. *Journal of Membrane Science*, 587, 117136.
16. Chung, A. J., & Hur, S. C. (2015). In *Micro- and Nanomanipulation Tools* (pp. 1–40).
17. Lenshof, A., & Laurell, T. (2010). Continuous separation of cells and particles in microfluidic systems. *Chemical Society Reviews*, 39(3), 1203–1217.
18. Huang, J. H., Jayaraman, A., & Ugaz, V. M. (2012). Enzymatic sculpting of nanoscale and microscale surface topographies. *Angewandte Chemie International Edition*, 51, 9619–9623.
19. Wyatt Shields Iv, C., Reyes, C. D., & López, G. P. (2015). Microfluidic cell sorting: A review of the advances in the separation of cells from debulking to rare cell isolation. *Lab on a Chip*, 15, 1230–1249.
20. Yoon, Y., Kim, S., Lee, J., Choi, J., Kim, R.-K., Lee, S.-J., Sul, O., & Lee, S.-B. (2016). Clogging-free microfluidics for continuous size-based separation of microparticles. *Scientific Reports*, 6, 26531.
21. Dijkshoorn, J. P., Schutyser, M. A. I., Wagterveld, R. M., Schroën, C. G. P. H., & Boom, R. M. (2017). A comparison of microfiltration and inertia-based microfluidics for large scale suspension separation. *Separation and Purification Technology*, 173, 86–92.
22. Lin, C.-K., Hsiao, Y.-Y., Nath, P., & Huang, J.-H. (2019). Aerosol delivery into small anatomical airway model through spontaneous engineered breathing. *Biomicrofluidics*, 13, 044109.
23. Hsieh, H.-L., Nath, P., & Huang, J.-H. (2019). Multistep fluidic control network toward the automated generation of organ-on-a-chip. *ACS Biomaterials Science & Engineering*, 5, 4852–4860.
24. Ottoboni, S., Price, C. J., Steven, C., Meehan, E., Barton, A., Firth, P., Mitchell, A., & Tahir, F. (2019). Development of a novel continuous filtration unit for pharmaceutical process development and manufacturing. *Journal of Pharmaceutical Sciences*, 108, 372–381.
25. Grilo, A. L., & Mantalaris, A. (2019). The increasingly human and profitable monoclonal antibody market. *Trends in Biotechnology*, 37, 9–16.
26. Bielser, J.-M., Wolf, M., Souquet, J., Broly, H., & Morbidelli, M. (2018). Perfusion mammalian cell culture for recombinant protein manufacturing – A critical review. *Biotechnology Advances*, 36, 1328–1340.

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